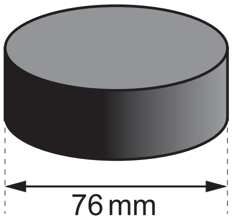
Examiner only

Option C – **The Physics of Sports**

8. This question is about the physics of the motion of an ice hockey puck which is a hard rubber disc of mass 0.17 kg and diameter 76 mm.



(a) When the disc is at room temperature, the coefficient of restitution between the puck and ice is 0.55.

(i) Explain what is meant by the statement *"the coefficient of restitution is 0.55"*. [2]

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(ii) When a puck is cooled from room temperature to 0 °C its coefficient of restitution is reduced by 30%. Calculate the bounce height of a puck at 0 °C when it is dropped from an initial height of 0.50 m on to ice. [4]

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(b) The image shows an ice hockey player taking a shot at goal.

(i) When striking the puck, the player changes its speed from $3\,m\,s^{-1}$ to $34\,m\,s^{-1}$ without changing its direction. The puck remains in contact with the hockey stick for 25 ms. Calculate the mean force exerted by the hockey stick on the puck. [2]

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(ii) The player aims for the top corner of the goal, which is at a height of 1.2 m. The initial angle of the puck's motion is 8° to the horizontal and its speed is $34\,m\,s^{-1}$. Determine whether or not the puck ever exceeds the height of the goal.
Ignore the effect of air resistance. [3]

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(iii) The spin rate of the puck at its maximum height is 14 revolutions per second. Calculate its **rotational** kinetic energy.
The moment of inertia of the puck is given by $I = \dfrac{mr^2}{2}$. [3]

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(iv) Wayne thinks that the answer to part *(b)*(iii) is actually the **total** kinetic energy of the puck at the maximum height. Determine whether Wayne is correct. [2]

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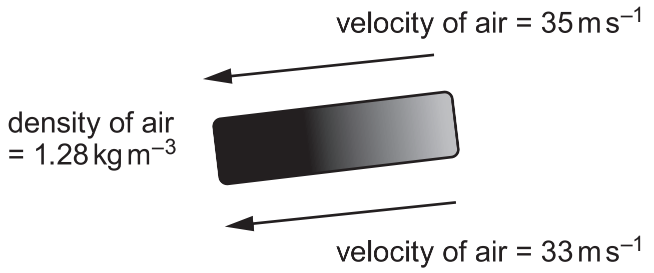
(v) During the shot at goal, the puck is moving to the right. The diagram below shows the velocity of the air relative to the puck. During its flight, the velocity of the air above the puck is greater than that below it creating lift.



Use the Bernoulli equation to calculate the lift force on the puck and show that this is small compared with its weight. [4]

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